

Econ 301 Homework – Model Specification

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1. Introduction

The purpose of this homework is to find a reasonable model specification for explaining the dependent variable, Y_i , using the explanatory variables $X_1, X_2, X_3, X_4, X_5, X_6$. Instead of choosing one final model directly, I estimate several alternative specifications and use the results of each estimation to guide the next step. The main focus is not only the final model, but also the reasoning process behind moving from one model to another.

In this analysis, I first examine the dataset through summary statistics and correlations. Then, I estimate a full regression model including all explanatory variables. After that, I check whether the model has possible violations of the Gauss-Markov assumptions, especially multicollinearity and heteroskedasticity. Based on these results, I estimate reduced models and compare them with the initial full model. Finally, I choose a preferred specification and report it with appropriate standard errors.

2. Data and Initial Analysis

The dataset contains 250 observations. The dependent variable is Y_i , and the possible explanatory variables are $X_1, X_2, X_3, X_4, X_5, X_6$. Since the variables are not given with specific economic names, the interpretation of the coefficients is mainly statistical. Therefore, I interpret each coefficient as the partial association between the relevant explanatory variable and Y_i , holding the other variables constant.

The table below reports the basic summary statistics for the dependent variable Y_i and the explanatory variables $X_1, X_2, X_3, X_4, X_5, X_6$.

Table 1. Summary Statistics from Stata

Variable	Obs	Mean	Std. dev.	Min	Max
y	250	14.14985	3.607249	2.202871	23.87586
x1	250	2.999344	1.000569	-.1532648	5.623152
x2	250	5.012234	.9894766	1.478342	8.031959
x3	250	1.954354	.9658916	-.770216	4.363613
x4	250	5.935431	1.044176	3.05523	8.683969
x5	250	3.999112	1.004237	1.173898	6.236786
x6	250	1.015585	.9335118	-1.92061	3.582236

The summary statistics show that the dataset contains 250 observations for all variables, so there are no missing values in the variables used in the analysis. The dependent variable Y_i has a mean of approximately 14.150 and a standard deviation of approximately 3.607. The explanatory variables also have variation, which makes them suitable for regression analysis. However, summary statistics alone do not show which variables should be included in the final model. For this reason, I next examine the correlations among the variables.

Table 2. Correlation Matrix from Stata

	y	x1	x2	x3	x4	x5	x6
y	1.0000						
x1	0.9769	1.0000					
x2	0.9579	0.9272	1.0000				
x3	-0.0447	0.1226	0.0857	1.0000			
x4	-0.0008	0.0009	0.0083	0.0556	1.0000		
x5	0.8827	0.8907	0.8804	0.1409	0.0477	1.0000	
x6	0.0772	0.0807	0.0720	0.0530	0.2266	0.1473	1.0000

The correlation matrix gives the first indication about possible relationships among the variables. The dependent variable Y_i is highly correlated with X_1 and X_2 , with correlations of approximately 0.977 and 0.958. This suggests that X_1 and X_2 may be important explanatory variables in the regression analysis. The variable X_5 is also strongly correlated with Y_i , with a correlation of approximately 0.883.

However, the correlation matrix also suggests a possible multicollinearity problem. The variables X_1 , X_2 , and X_5 are strongly correlated with each other. For example, the correlation between X_1 and X_2 is approximately 0.927. The correlations between X_1 and X_5 , and between X_2 and X_5 , are also high. This does not prove that the model is misspecified, but it suggests that multicollinearity should be checked more formally after estimating the first regression model.

3. Full Model Estimation

I first estimate the unrestricted model including all six explanatory variables. This model is used as the benchmark specification because it allows each variable to explain Y_i while controlling for all other available variables. The estimated full model is:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + \beta_5 X_{5i} + \beta_6 X_{6i} + E_i$$

Equation 1. The unrestricted model

The results of the unrestricted model are reported in Table 3. This table shows the estimated coefficients, standard errors, t-statistics, and p-values for each explanatory variable.

Table 3. Unrestricted Model Regression Results from Stata

Source	SS	df	MS	Number of obs	=	250
Model	3233.80518	6	538.96753	F(6, 243)	=	20976.68
Residual	6.24355722	243	.025693651	Prob > F	=	0.0000
				R-squared	=	0.9981
				Adj R-squared	=	0.9980
Total	3240.04874	249	13.0122439	Root MSE	=	.16029

y	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
x1	2.470638	.0299806	82.41	0.000	2.411583	2.529693
x2	1.251416	.0291361	42.95	0.000	1.194025	1.308808
x3	-.5879066	.010694	-54.98	0.000	-.6089714	-.5668418
x4	.0122713	.0100224	1.22	0.222	-.0074706	.0320132
x5	-.0314851	.0240219	-1.31	0.191	-.0788028	.0158325
x6	.0232101	.0113591	2.04	0.042	.0008352	.045585
_cons	1.645648	.0918322	17.92	0.000	1.46476	1.826537

The unrestricted model has a very high R-squared of 0.998 and an adjusted R-squared of 0.998. This means that the model explains almost all of the variation in Y_i . The F-statistic is also very large, which suggests that the explanatory variables are jointly significant in explaining Y_i . However, the individual coefficient results show that not every variable contributes equally to the model.

The p-values of X_1 , X_2 , and X_3 are very close to zero (rounded up to zero by Stata), so these variables are clearly statistically significant. This means that they appear to be strongly related to Y_i when the other variables are kept fixed. The coefficients of X_1 and X_2 are positive, meaning that increases in these variables are associated with increases in Y_i , holding the other variables constant. The coefficient of X_3 is negative, meaning that an increase in X_3 is associated with a decrease in Y_i , holding the other variables constant.

The coefficient of X_6 is also statistically significant at the 5% level because its p-value is 0.042. However, its coefficient is small compared with the coefficients of X_1 , X_2 , and X_3 . In contrast, X_4 and X_5 are not statistically significant in the unrestricted model. Their p-values are 0.222 and 0.191, which are both above the 5% significance level. This suggests that, after keeping other variables fixed, X_4 and X_5 do not add strong explanatory power to the model. This result is especially important for X_5 , because the earlier correlation matrix showed that X_5 is strongly correlated with Y_i . Once X_1 and X_2 are included in the regression, however, X_5 no longer appears statistically significant. Therefore, the next step is to check whether this is connected to multicollinearity and whether X_4 and X_5 should be omitted from the next specification.

Since the correlation matrix showed high correlations among some explanatory variables, I next check multicollinearity using VIF values. This is useful because multicollinearity refers to strong, but not perfect, correlation among explanatory variables. In this case, the earlier correlation matrix suggested that X₁, X₂, and X₅ may be strongly related to each other. The VIF for variable X_j is calculated as:

$$VIF_j = 1 / (1 - R_j^2)$$

Equation 2. Variance Inflation Estimator¹

Where R_j^2 is obtained from an auxiliary regression of X_j on the other explanatory variables. Therefore, a higher VIF means that the variable can be explained more strongly by the other explanatory variables. This makes it harder to separate its individual effect in the main regression. The VIF results for the unrestricted model are reported in Table 4.

Table 4. VIF Results for the Unrestricted Model from Stata

Variable	VIF	1/VIF
x1	8.72	0.114670
x2	8.05	0.124152
x5	5.64	0.177313
x6	1.09	0.917692
x4	1.06	0.942181
x3	1.03	0.967137
Mean VIF	4.27	

The VIF results support the concern raised from the correlation matrix. The highest VIF values belong to X₁, X₂, and X₅. Their VIF values are 8.72, 8.05, and 5.64, respectively. These values do not indicate perfect multicollinearity, but they are high enough to suggest a possible multicollinearity problem.

This matters because multicollinearity can make it harder to separate the individual effects of explanatory variables that move together. In this model, X₅ is highly correlated with X₁ and X₂, and it is also statistically insignificant in the unrestricted regression. Therefore, X₅ becomes a candidate for removal in the next specification. X₄ also has a statistically insignificant coefficient, although its VIF value is low. For this reason, I next test whether X₄ and X₅ can be jointly omitted from the model.

The next step is to test whether X₄ and X₅ can be jointly omitted from the model. This is necessary because both variables are individually insignificant in the unrestricted regression. The joint test examines the null hypothesis that B₄ = B₅ = 0.

Table 5. Joint Test Results from Stata for Omitting X₄ and X₅

Test	Null Hypothesis	F-statistic	p-value	Decision
Joint significance test	B ₄ = B ₅ = 0	1.52	0.2213	Fail to reject H ₀

The next step is to test whether X₄ and X₅ can be jointly omitted from the model. This is necessary because both variables are individually insignificant in the unrestricted regression. The joint test examines the null hypothesis that B₄ = B₅ = 0. The test gives an F-statistic of 1.52. Using the F-table² at the 5% significance level, with 2 numerator degrees of freedom and 243 denominator degrees of freedom, the critical value is about 3.03. Since 1.52 is smaller than 3.03, I fail to reject the null

¹ From the Lectures of Econ 301

² Check Appendix for the F-Table

hypothesis. The p-value of 0.2213 also supports the same conclusion because it is above 0.05. Therefore, X_4 and X_5 are not jointly significant, and omitting them from the next specification is reasonable.

4. Restricted Model Estimation

Omitting variables can create bias if the omitted variables actually belong in the true model and are related to the included explanatory variables. Therefore, omission should not be used mechanically as a cure for multicollinearity. This point can be understood through the mean squared error approach.

$$MSE = E [\hat{\beta} - \beta]^2 = Var(\beta) + Bias(\beta)^2$$

Equation 3. Mean Squared Error³

Omission can also have benefits. If variables do not add meaningful explanatory power, keeping them may make the model unnecessarily complicated and may increase the variance of the coefficient estimates. In this case, X4 and X5 are not omitted only because of multicollinearity. They are also individually insignificant, and the joint test fails to reject $B4 = B5 = 0$.

After this, I estimate the restricted model by omitting X4 and X5 from the unrestricted model. The restricted model is:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_6 X_{6i} + E_i^*$$

Equation 4. The restricted model

Here, E_i^* represents the new error term in the restricted model. Since X4 and X5 are omitted, any remaining influence of these variables would be absorbed into this new error term. This is why omission can create bias if the omitted variables are actually relevant. However, in this case, the previous joint test failed to reject $B4=B5=0$, so estimating the restricted model is reasonable.

The results of the restricted model are reported in Table 6. This table shows whether the remaining variables continue to explain Y_i after X_4 and X_5 are omitted.

Table 6. Restricted Model Regression Results from Stata

Source	SS	df	MS	Number of obs	=	250
Model	3233.7272	4	808.4318	F(4, 245)	=	31331.90
Residual	6.32153793	245	.025802196	Prob > F	=	0.0000
				R-squared	=	0.9980
				Adj R-squared	=	0.9980
Total	3240.04874	249	13.0122439	Root MSE	=	.16063

y	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
x1	2.453391	.0273524	89.70	0.000	2.399515	2.507267
x2	1.239581	.0275377	45.01	0.000	1.18534	1.293822
x3	-.5885818	.0106593	-55.22	0.000	-.6095773	-.5675863
x6	.0237633	.0109507	2.17	0.031	.0021937	.0453328
_cons	1.704378	.0737004	23.13	0.000	1.559211	1.849545

The results in Table 6 show that the restricted model remains very strong after omitting X4 and X5. The R-squared and adjusted R-squared are both approximately 0.998, which is almost the same as in the unrestricted model. The overall F-statistic also increases from 20976.68 in the unrestricted model to 31331.90 in the restricted model. Although these two F-statistics do not test exactly the same null hypothesis, this comparison shows that the remaining variables are jointly very strong after omitting X4 and X5.

The coefficient results also support this interpretation. After omitting X4 and X5, all remaining explanatory variables are statistically significant. The p-values of X1, X2, and X3 are 0.000, and the p-value of X6 is 0.031. Therefore, the restricted model is simpler and does not perform worse than the

³ From the Lectures of Econ 301

unrestricted model. Since X4 and X5 were individually and jointly insignificant, the restricted model is a reasonable specification.

After estimating the restricted model, I check the VIF values again to see whether omitting X5 reduced the multicollinearity concern identified in the unrestricted model.

Table 7.VIF Results for the Restricted Model from Stata

Variable	VIF	1/VIF
x1	7.23	0.138348
x2	7.16	0.139569
x3	1.02	0.977563
x6	1.01	0.991591
Mean VIF	4.11	

The results in Table 7 show that omitting X5 reduces the multicollinearity concern, but does not completely remove it. In the unrestricted model, the VIF values of X1 and X2 were 8.72 and 8.05. In the restricted model, they fall to 7.23 and 7.16. The mean VIF also decreases from 4.27 to 4.11. This suggests that omitting X5 improves the multicollinearity problem to some extent.

However, X1 and X2 still have relatively high VIF values because they remain strongly related to each other. I do not omit either X1 or X2 only to reduce VIF, because both variables remain strongly statistically significant in the restricted model. Omitting one of them could create a more serious omitted variable bias problem. Therefore, I keep X1 and X2 in the restricted model and move to the next diagnostic issue: heteroskedasticity.

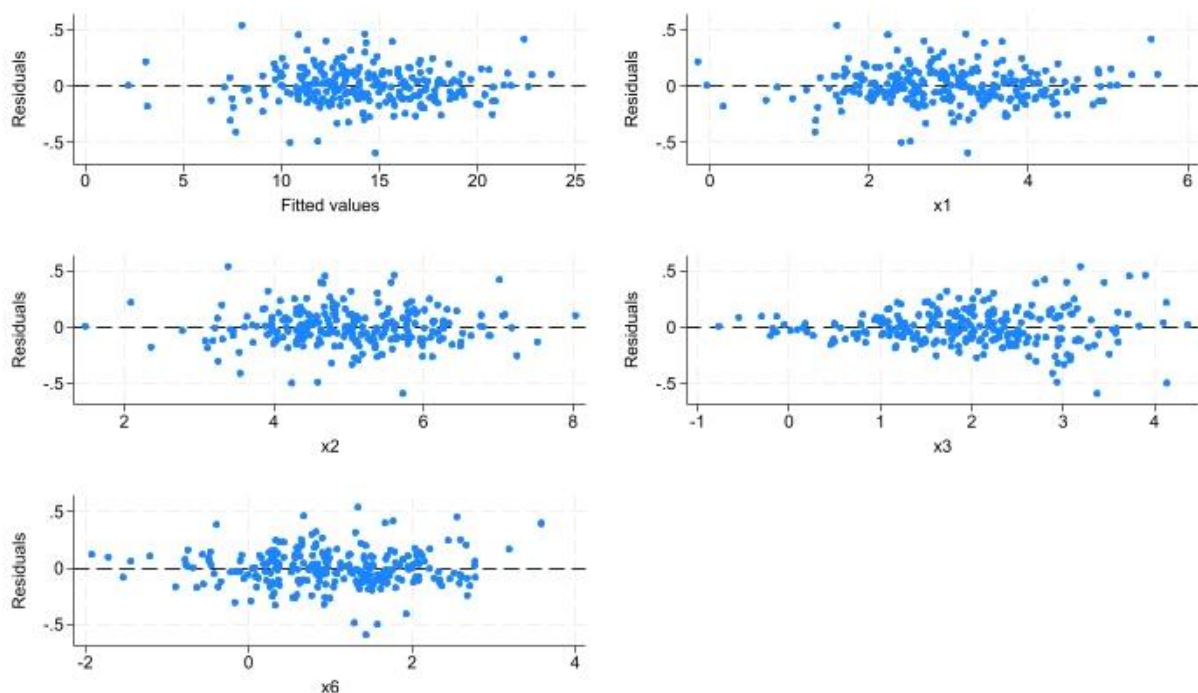
5. Heteroskedasticity

After choosing the restricted model, I next check whether the model has a heteroskedasticity problem. Heteroskedasticity means that the variance of the error term is not constant across observations. This matters because the usual OLS variance formula may become unreliable, which also affects standard errors, t-statistics, and p-values.

There are two possible cases. If the source of heteroskedasticity is known or predictable from the data, then a weighted least squares approach can be used by transforming the model with appropriate weights. However, if the source of heteroskedasticity is unknown, then it is more appropriate to keep the OLS coefficients and use robust variances. Therefore, in this section, I first test whether heteroskedasticity exists in the restricted model. Then, depending on the results, I decide whether robust variances are necessary for reliable statistical inference.

To begin the heteroskedasticity analysis, I first examine the residual plots from the restricted model. These plots are useful because heteroskedasticity may appear visually as a changing spread of residuals across fitted values or explanatory variables. Therefore, before using formal tests, I check whether the residuals seem to have a constant or changing variance.

Figure 1. Residual Plots from Stata for the Restricted Model



The residual plots in Figure 1 provide a visual check for heteroskedasticity in the restricted model. The residuals are generally centered around zero, but their spread does not look perfectly constant across the fitted values and explanatory variables. This visual pattern suggests a possible heteroskedasticity problem. However, since visual inspection alone is not enough to reach a final conclusion, I next conduct formal heteroskedasticity tests.

Table 8. Heteroskedasticity Test Results from Stata

Test	Null Hypothesis (H0)	Test Statistic	p-value	Decision
Breusch-Pagan	Constant variance	chi2(1) = 4.50	0.0340	Reject H0
White Test	Constant variance	chi2(14) = 51.28	0.0000037	Reject H0

The Breusch-Pagan test gives a p-value of 0.0340, which is below the 5% significance level. Therefore, I reject the null hypothesis of constant variance. The White test also rejects the null hypothesis, with a p-value of approximately 0.0000037. Since both tests reject constant variance, there is evidence of heteroskedasticity in the restricted model.

After finding evidence of heteroskedasticity, the next question is whether the origin of heteroskedasticity is known or unknown. If the origin is known, weighted least squares can be considered. If the origin is unknown, then robust variances are more appropriate. For this reason, I examine the squared residual plots and estimate an auxiliary regression of squared residuals on the explanatory variables.

Figure 2. Squared Residual Plots from Stata Restricted Model

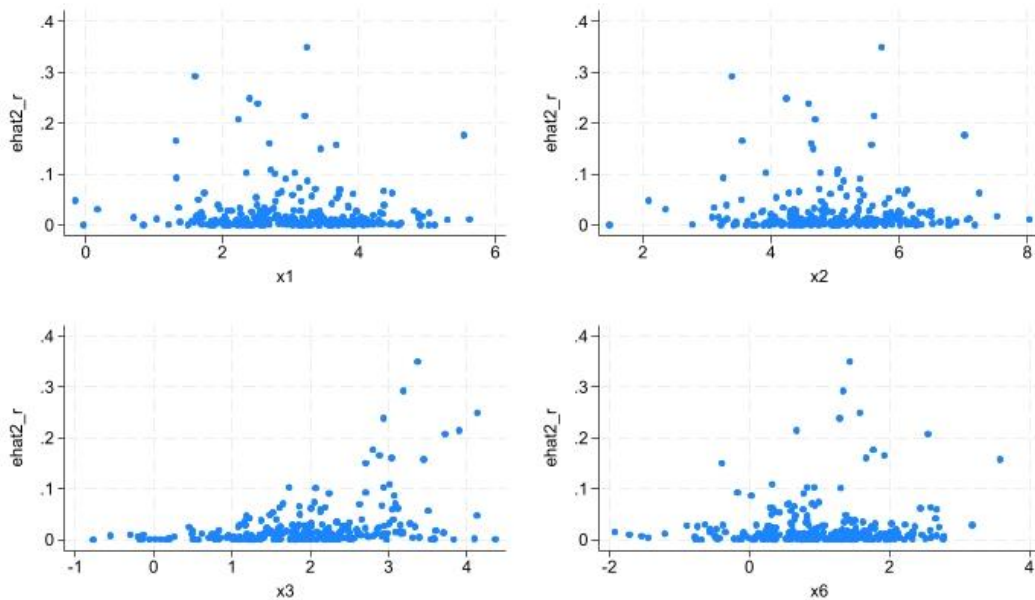


Table 9. Auxiliary Regression Results from Stata for the Restricted Model

Source	SS	df	MS	Number of obs	=	250
Model	.08624147	4	.021560367	F(4, 245)	=	10.94
Residual	.482831901	245	.001970742	Prob > F	=	0.0000
				R-squared	=	0.1515
				Adj R-squared	=	0.1377
Total	.569073371	249	.002285435	Root MSE	=	.04439

e_hat2_r	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
x1	-.0066008	.0075593	-0.87	0.383	-.0214903	.0082887
x2	.0023169	.0076105	0.30	0.761	-.0126735	.0173073
x3	.0188492	.0029459	6.40	0.000	.0130467	.0246517
x6	.0037479	.0030264	1.24	0.217	-.0022133	.009709
_cons	-.0071729	.0203684	-0.35	0.725	-.0472923	.0329466

The squared residual plots and the auxiliary regression show that the origin of heteroskedasticity is related to X3. In Figure 2, the squared residuals show a clearer increasing pattern with X3 than with the other explanatory variables. Table 9 supports this visual pattern. The auxiliary regression is jointly significant, with a p-value of 0.0000. Among the explanatory variables, only X3 is statistically significant, with a positive coefficient and a p-value of 0.000. This means that squared residuals increase as X3 increases. Therefore, the origin of heteroskedasticity is treated as known and related to X3.

Since the origin of heteroskedasticity is related to X3, I next consider a weighted least squares model based on X3. This allows me to check whether WLS improves the model compared with the restricted OLS specification.

6. Weighted Least Squares

Since the origin of heteroskedasticity is treated as known and related to X3, I next estimate a weighted least squares model based on X3. For this WLS transformation, I assume that the variance of the restricted model error term is proportional to X_{3i}^2 :

$$Var(E_i^*/X) = \sigma^2 X_{3i}^2$$

Under this assumption, dividing the model by $X_{3i}X_{3i}$ gives a transformed error term with constant variance:

$$Var(E_i^*/X_{3i}) = \sigma^2$$

Therefore, I transform the restricted model by dividing each variable by $X_{3i}X_{3i}$. The transformed variables are:

$$Y_i^* = Y_i / X_{3i}, X_{1i}^* = X_{1i} / X_{3i}, X_{2i}^* = X_{2i} / X_{3i}, X_{3i}^* = X_{3i} / X_{3i} = 1,$$

$$X_{6i}^* = X_{6i} / X_{3i}, C_i^* = 1 / X_{3i}.$$

So, the transformed WLS model is:

$$Y_i^* = \beta_0 C_i^* + \beta_1 X_{1i}^* + \beta_2 X_{2i}^* + \beta_3 + \beta_6 X_{6i}^* + E_i^*$$

Equation 5. The WLS restricted model

Table 10. WLS Regression Results from Stata

Source	SS	df	MS	Number of obs	=	250
Model	512299.725	5	102459.945	F(5, 245)	>	99999.00
Residual	2.19093856	245	.008942606	Prob > F	=	0.0000
				R-squared	=	1.0000
				Adj R-squared	=	1.0000
Total	512301.916	250	2049.20766	Root MSE	=	.09457

ystar	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
x1star	2.445037	.0096361	253.74	0.000	2.426057	2.464017
x2star	1.261226	.0078202	161.28	0.000	1.245823	1.27663
x6star	-.0157014	.0048834	-3.22	0.001	-.0253202	-.0060827
constantstar	1.670606	.0136136	122.72	0.000	1.643792	1.697421
x3term	-.5991464	.0061367	-97.63	0.000	-.6112337	-.587059

The WLS results are consistent with the main findings of the restricted OLS model. X1 and X2 remain positive and statistically significant, while X3 remains negative and statistically significant. Therefore, the main interpretation of the model does not change after the WLS transformation. X6 becomes negative and statistically significant, but its coefficient is small, so it should be interpreted cautiously.

Since the heteroskedasticity tests rejected constant variance and the auxiliary regression suggested that the problem is related to X3, the WLS transformation based on X3 is a reasonable correction. For this reason, I use the WLS model as the preferred final specification.

7. The Final Model

After estimating the unrestricted model, the restricted model, and the WLS model, I choose the WLS version of the restricted model as the final specification. The unrestricted model showed that X4 and X5 were not statistically significant. The joint test also failed to reject the null hypothesis that these two variables are jointly equal to zero. Therefore, omitting X4 and X5 was reasonable.

The restricted model kept X1, X2, X3, and X6, and all of these variables were statistically significant. However, the Breusch-Pagan and White tests showed evidence of heteroskedasticity. Since the auxiliary regression suggested that this problem is related to X3, I corrected the restricted model using WLS. Therefore, the final preferred model is:

$$Y_i^* = \beta_0 C_i^* + \beta_1 X_{1i}^* + \beta_2 X_{2i}^* + \beta_3 + \beta_6 X_{6i}^* + E_i^*$$

Based on the WLS results from table 10, the estimated final equations are:

$$\hat{Y}_i^* = 1.6706 C_i^* + 2.4450 X_{1i}^* + 1.2612 X_{2i}^* - 0.5991 - 0.0157 X_{6i}^*$$

$$\hat{Y}_i = 1.6706 + 2.4450 X_{1i} + 1.2612 X_{2i} - 0.5991 X_{3i} - 0.0157 X_{6i}$$

These equations summarize the final WLS specification and will be used as the preferred final model.

8. Conclusion

This homework followed a step-by-step model specification process. I first estimated the unrestricted model with all six explanatory variables. Although the model had very high explanatory power, X4 and X5 were not statistically significant. The VIF results also showed that X5 was strongly related to X1 and X2, which created a multicollinearity concern.

I then tested whether X4 and X5 could be omitted. Since the joint test failed to reject the null hypothesis, I estimated a restricted model with X1, X2, X3, and X6. This model remained strong, but the Breusch-Pagan and White tests showed evidence of heteroskedasticity, meaning that the constant variance assumption was violated. Therefore, one of the Gauss-Markov assumptions was not satisfied in the restricted OLS model.

To correct this heteroskedasticity problem, I estimated a WLS model based on the assumption that the error variance is proportional to X_{3i}^2 , that is, $Var(E_i|X) = \sigma^2 X_{3i}^2$. The WLS model keeps the important variables from the restricted model and responds to the detected Gauss-Markov violation by transforming the model with weights based on X3. The WLS results also show a very good fit, since the explanatory variables remain statistically significant and the model has very high explanatory power after the transformation. Therefore, I choose the WLS model as the final preferred specification.

Overall, the final model suggests that X1 and X2 are positively related to Y, while X3 is negatively related to Y. The coefficient of X6 is negative in the WLS model, but its size is small and its sign changed compared with the restricted OLS model. Therefore, X6 appears less stable than X1, X2, and X3, so its effect should be interpreted cautiously.

9. Appendix A. F Table

The joint test for omitting X4 and X5 uses the following null hypothesis:

$$H_0: \beta_4 = \beta_5 = 0$$

Since two restrictions are tested, the numerator degrees of freedom is:

$$v_1 = 2$$

The unrestricted model has 250 observations and 7 estimated parameters, including the constant. Therefore, the denominator degrees of freedom is:

$$v_2 = n - k = 250 - 7 = 243$$

Denominator df v_2 / Numerator df v_1	1	2	3	4	5
200	3.89	3.04	2.65	2.42	2.26
250	3.88	3.03	2.64	2.41	2.25
300	3.87	3.03	2.63	2.40	2.24

Source: Wooldridge, J. M. (2020). *Introductory Econometrics: A Modern Approach (7th ed.)*. Cengage Learning.

From the F-Table, the critical value is: $F_{crit} \approx 3.03$

10. Appendix B. Stata Do-File Codes

```
cls
clear all
set more off
import excel "Econ301 Homework Dataset (1)", firstrow clear
describe
summarize y x1 x2 x3 x4 x5 x6
correlate y x1 x2 x3 x4 x5 x6
regress y x1 x2 x3 x4 x5 x6
*Multicollinearity
vif
test x4 x5
regress y x1 x2 x3 x6
vif
*Heteroscedasticity
predict yhat_r
predict ehat_r, resid
scatter ehat_r yhat_r, yline(0) name(g1, replace)
scatter ehat_r x1, yline(0) name(g2, replace)
```

```
scatter ehat_r x2, yline(0) name(g3, replace)
scatter ehat_r x3, yline(0) name(g4, replace)
scatter ehat_r x6, yline(0) name(g5, replace)
graph combine g1 g2 g3 g4 g5, cols(2) name(graph1)
```

```
hettest
```

```
ssc install whitetst, replace
```

```
whitetst
```

*Origin of the Heteroscedasticity

```
gen ehat2_r = ehat_r^2
scatter ehat2_r x1, name(s1, replace)
scatter ehat2_r x2, name(s2, replace)
scatter ehat2_r x3, name(s3, replace)
scatter ehat2_r x6, name(s4, replace)
graph combine s1 s2 s3 s4, cols(2) name(graph2)
regress ehat2_r x1 x2 x3 x6
```

*WLS

```
gen ystar = y / x3
gen x1star = x1 / x3
gen x2star = x2 / x3
gen x6star = x6 / x3
gen constantstar = 1 / x3
gen x3term = 1
regress ystar x1star x2star x6star constantstar x3term, noconstant
predict wyhat
predict wehat, resid
scatter wehat wyhat, yline(0) name(wg1, replace)
graph combine g1 wg1, cols(2) name(graph3, replace)
```